



Supplementary Materials for

Predicting temperature mortality and selection in natural *Drosophila* populations

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Published 4 September 2020, *Science* **369**, page (2020)
DOI: 10.1126/science.aba9287

This PDF file includes:

Materials and Methods
Figs. S1 to S5
Table S1
References

Other Supplementary Material for this manuscript includes the following:
(available at science.sciencemag.org/content/369/6508/1242/suppl/DC1)

MDAR Reproducibility Checklist

Supplementary Online Materials

Page

2.	Table of contents
3.	Materials and Methods
5.	Analytical model
6.	<i>Temperature ramp</i>
7.	<i>Recovery</i>
8.	Implementation of the model
8.	<i>Fitting the model to static data</i>
9.	<i>Predicting mortality at variable temperatures</i>
10.	<i>Worked examples</i>
13.	Validation with empirical datasets
13.	<i>Static assays</i>
13.	<i>Figure S1 – Castañeda et al. (24) data set: warm-acclimated flies</i>
13.	<i>Figure S2 – Castañeda et al. (24) data set: cold-acclimated flies</i>
14.	<i>Figure S3 – Jørgensen et al. (17) data set</i>
16.	<i>Dynamic assays</i>
16.	<i>Figure S4 – Predicting mortality in dynamic (ramping) assays</i>
20.	Thermal mortality and global warming
20.	<i>Table S1 – Historical temperature records for 1984–1991 and 2014–2018</i>
21.	<i>Figure S5 – Shift in mortality rates during 30 years of global warming</i>

Materials and Methods

We provide the detailed analytical model to estimate the survival probability function $S(\tau)$ under a variable thermal regime in “Supplementary text: Analytical model”, and implemented the model in a script made *ad hoc* in R (<https://cran.r-project.org>). How the model is applied to empirical data is described in “Supplementary text: Implementation of the model”, and the R functions described in this section are available at DRYAD (see main text).

Starting with data on knockdown or death times obtained at different constant stressful temperatures $T_1, T_2, \dots$, which describe $S_{T_1}, S_{T_2}, \dots$ (Fig. 1), the script performs two separate procedures (“Supplementary text: Fitting the model to static data”; “Supplementary text: Predicting mortality in ramping experiments”). In the first place, it calculates the general survival probability function $S(\tau)$ by centering all survival probability curves at the average temperature $\bar{T}$, effectively pooling all the empirical data onto a single curve, and tests the general assumption that this curve adequately describes survival curves obtained at different temperatures when it is shifted by $10^{\frac{\Delta T}{z}}$. The test is accomplished with a linear regression of the fitted $\log_{10}(\tau)$ based on $S(\tau)$ against the empirical estimates described by $S_{T_1}, S_{T_2}, \dots$. For the *Drosophila* dataset analyzed here, which constitutes the largest and most highly resolved dataset describing thermal tolerance across multiple constant temperatures (17), these diagnostic regressions between fitted and empirical $\log_{10}(\tau)$ resulted in r^2 ranging between 0.946 and 0.981 (Fig. 2a, “Supplementary text: Predicting mortality in ramping experiments”). In the second place, having ensured that $S(\tau)$ and z can accurately encapsulate the variation in empirical curves, the script builds a thermal tolerance landscape describing survival curves at different temperatures with a 0.1 °C resolution and estimates the predicted survival probability curve under a variable thermal regime with temperatures every 1-min specified by the user. We validated this script against survival curves reported for 11 *Drosophila* species, which were measured under three ramping conditions (17), resulting in a total of 33 separate survival curves (“Supplementary text: Predicting mortality in ramping experiments”). Validation was performed with a linear regression of the theoretically predicted lethal times and temperatures versus empirical values (Fig. 2b, c).

To estimate thermal mortality in the field, we combined survival curves obtained for a *D. subobscura* outbred population collected in Santiago (23) with ambient temperature records from 2014 to 2018 obtained in an hourly basis from the closest weather station (La Platina, <http://agromet.inia.cl/>). Hourly temperatures were averaged across years to minimize the impact of extreme events, removing February 29 of leap year 2016 for simplicity, and records with 1 min resolution were estimated with non-linear interpolation between consecutive hours ('fmm' spline function in R). To reconstruct hourly temperatures for 1984 – 1991, we employed records for 2014 – 2018 and removed long-term warming trends apparent in average daily maxima obtained for both periods (Table S1). The general impact of warming temperatures was quantified by contrasting monthly daily maxima between these periods.

Predicted mortality was calculated on a daily basis starting with a survival probability $s = 1$, which inherently assumes that at night flies can recover from the heat stress experienced during the day. The cumulative mortality was calculated as $1 - \prod s_i$, where s_i corresponds to the probability to survive each day (Eqn. S9). Predicted mortality was contrasted against estimates of abundance of *D. subobscura* recorded in Santiago for 8 consecutive years (24), which were expressed in relative scale with yearly maxima standardized to 1 for comparative purposes (Fig. 3c). Briefly, in this study flies were collected monthly from August 1984 to December 1991 (note there is a typo in material and methods of ref. 24 because the last sampling was in December 1991, not December 1990) by net-sweeping over fermented banana baits at evening around sunset (24).

Analytical model

Here we present the method to obtain survival probability curves under a variable thermal regime from empirical estimates obtained at different constant temperatures. The two following subsections describe the analytical procedures detailed schematically in Fig 1.

The thermal shift (Fig. 1 from the left panel to the middle one)

As $\log(\tau)$ is a linear function of T (Fig. 1a) we can get the following equation:

$$\log(\tau(T)) = -\frac{1}{z} \cdot T + D(S) \Rightarrow \tau(T) = d(S) \cdot 10^{-\frac{T}{z}}, \quad (\text{S1})$$

where $-\frac{1}{z}$ is the (logarithmic) slope corresponding of the thermal death time (TDT) curves, and $d(S) = 10^{D(S)}$ the vertical intercept. Note that the slope is independent of survival rate S ; the iso- S line are parallel in the logarithmic plot. From that we have a simple relation that makes connection between thermal death times and temperatures for any given survival rate

$$\tau_2(T_2) = \tau_1(T_1) \cdot 10^{-\frac{1}{z}(T_2 - T_1)} \quad (\text{for any given } S). \quad (\text{S2})$$

As for any survival rate S there are two corresponding T, τ pairs to satisfy $S_{T_1}(\tau_1) = S_{T_2}(\tau_2)$, by using the previous equation we get

$$S_{T_1}(\tau_1) = S_{T_2}(10^{-\frac{1}{z}(T_2 - T_1)} \tau_1). \quad (\text{S3})$$

As this is true for any τ we can omit the index and get

$$S_{T_1}(\tau) = S_{T_2}(10^{-\frac{1}{z}(T_2 - T_1)} \tau), \quad (\text{S4})$$

which defines a relation between the $S_T(\tau)$ curves at different temperatures (cf. Fig. 1b). Note that Eqn. (S4) does not define the $S_T(\tau)$ in a unique way, hence the proposed method is general and can accommodate any survival probability function obtained analytically or numerically (thermal death time isoclines will be closer or farther apart depending on the

overall shape of $S_T(\tau)$, but they will always remain parallel). In our analyses we estimate $S_T(\tau)$ numerically (see ‘*Fitting the model to static data*’ below).

Temperature ramp

As, in the general case, the survival rate (S) is a function of time (τ) and temperature (T), a small change in S can be expressed in the following way

$$dS_{T(\tau)}(\tau) = \frac{\partial S_{T(\tau)}(\tau)}{\partial \tau} d\tau + \frac{\partial S_{T(\tau)}(\tau)}{\partial T} dT. \quad (S5)$$

Because temperature is also a function of time ($T = T(\tau)$) the total derivative can be expressed as

$$dS_{T(\tau)}(\tau) = \left[\frac{\partial S_{T(\tau)}(\tau)}{\partial \tau} + \frac{\partial S_{T(\tau)}(\tau)}{\partial T} \frac{\partial T(\tau)}{\partial \tau} \right] d\tau, \quad (S6)$$

where $\frac{\partial T(\tau)}{\partial \tau}$ is the temperature gradient. If there is an analytical form for $S_T(\tau)$ the survival rate at any given time τ' can be computed by integrating the total derivative as

$$S(\tau') = 1 + \int_0^{\tau'} dS_{T(\tau)}(\tau), \quad (S7)$$

where we use $S_T(0) = 1$ for any T . Otherwise, we can compute the survival function from the numerical derivatives $S'(k \cdot \Delta\tau)$, or the empirical slopes of decreasing survival at discrete time series $k \cdot \Delta\tau$, that can be approximated from the data at temperature $T(k \cdot \Delta\tau)$. In this case the survival function can be approximated by numerical integration (c.f. Eqn. S7) as

$$S(k \cdot \Delta\tau) \approx 1 + \sum_k S'(k \cdot \Delta\tau) \Big|_{T(k \cdot \Delta\tau)} \Delta\tau. \quad (S8)$$

To summarize, Eqn. S1 to S7 demonstrate that survival probability curves under variable thermal conditions can be predicted from curves estimated at different constant temperatures. However, because the function $S_T(\tau)$ is not defined for most curves obtained with empirical data, we rely on numerical approximations to implement the model. The numerical

implementation of Eqn. S8 to obtain survival under variable conditions is logistically complex and described below (see ‘*Predicting mortality in ramping experiments*’ below).

Recovery

As currently implemented, our model estimates the thermal mortality within a relatively short period (e.g., the model was validated with ramping experiments lasting < 12 h) and does not explicitly include a recovery process. To estimate the cumulative mortality following days or weeks of exposure to stressful temperatures, a full recovery can be modelled by ‘resetting’ the survival curve to $S_T(0) = 1$ at the beginning of each 24 h cycle and estimating the fraction of the population S that survived the whole period as

$$S = \prod_{d=1}^n S_d \quad (\text{S9})$$

where S_d correspond to the survival fraction at the end of each day. Note that without this resetting procedure, S over long periods would reflect the cumulative impact of multiple days of thermal stress with no recovery during the night (i.e., carryover effects). Diagnostic analyses contrasting the cumulative mortality predicted with carryover effects versus the relative abundance of *D. subobscura* in the field show that this procedure highly overestimates mortality over multiple days or weeks (E.L. Rezende, pers. obs.), which indicates that flies must recover to some degree from a thermal stress during the night. Partial recovery may also be implemented by resetting the survival curve to $S_T(0) = 1 - \alpha S_{d-1}$, which assumes that a fraction α of survivors at day $d - 1$ do not fully recover and amplifies cumulative mortality due to carryover effects ($\alpha = 0$ and 1 correspond to full and no recovery, respectively). While this approach does not explicitly model recovery in time, empirical measurements of the survivors’ recovery dynamics across different thermal stress and recovery scenarios in *Drosophila* or other species are currently unavailable to parameterize such a model. Needless to say, with the development of a dynamic model to study thermal stress and mortality, studying the dynamics of recovery and its potential impact on demography and population resilience becomes increasingly important.

Implementation of the model

Fitting the model to static data

We then validated the predictions of the analytical model described above against empirical measurements, employing ordinary least-squares regression. This procedure collapses survival probability curves obtained at different T into a single curve $S(\tau)$ that can be shifted by a factor of z^{-1} depending on the intensity of the thermal stress (note that in the thermal treatment literature z corresponds to a thermal sensitivity parameter that describes the ΔT required to change τ in one order of magnitude). The step-by-step procedure:

- i. Calculate the survival curve at each T from empirical measurements of τ . This is done by sorting τ from smallest-to-largest and estimating $S_{T_i}(\tau) = \#dead/total$ at τ .
- ii. Calculate z with linear regression of $\log_{10} \tau$ versus T .
- iii. Calculate average $\bar{T}$.
- iv. Shift empirical survival probability curves measured at each T to obtain multiple overlapping survival curves at $\bar{T}$. This is accomplished employing
$$S_{T_1}(\tau) = S_{T_2} \left(10^{-\frac{(T_2 - T_1)}{z}} \tau \right).$$
- v. Build expected $S(\tau)$ by pooling all empirical estimates into a single probability survival curve, repeating the procedure described in Step i.
- vi. Map fitted $S(\tau)$ back onto the measurement temperatures as described in Step iv.
- vii. Compare fitted $S(\tau)$ against $S_{T_i}(\tau)$ employing an ordinary least-squares linear regression.

These steps are performed with the *tolerance.landscape* function built *ad hoc*, that reconstructs $S_T(\tau)$ at the average temperature $\bar{T}$ employing as input data (1) the measurement temperature and (2) the empirical knockdown or lethal time τ for each individual (note that the shape of $S_T(\tau)$ is determined by the distribution of τ). The script does not require $S_T(\tau)$ to be defined in analytical form, and instead describes the decrease in survival probability from 1 to 0 by dividing the empirical curve in 1000 time bins of variable length

that correspond to a drop in survival of 0.1% each. The reason to work with constant bins of survival probability in the 'y-axis' and not time in the 'x-axis' is because time is highly variable across temperatures (Eq. 1), whereas survival probability is always bounded between 1 and 0.

Usage:

```
tl <- tolerance.landscape(temperature,time)
```

Output:

tl\$ctmax: Temperature resulting in 50% mortality after 1 min exposure (see ref. 16).

tl\$z: Thermal sensitivity parameter z .

tl\$ta.mn: Average temperature $\bar{T}$ for which $S_T(\tau)$ is calculated.

tl\$S: Two-column data frame with 1000 survival and time bins describing $S_T(\tau)$.

tl\$time.obs.pred: Empirical versus fitted lethal times for diagnostics.

tl\$rsq: r^2 of the linear regression between empirical versus fitted lethal times.

Predicting mortality at variable temperatures

The function $S_T(\tau)$ obtained from the procedure above is subsequently used to predict survival probability under variable thermal conditions as described in Eqn. S8. This is carried out with function *dynamic.landscape*, which requires as input data (1) the body temperature with 1-min resolution (= air temperature in our analyses, which may not correspond to the thermal microhabitats for aquatic species or lineages that shelter underground) and (2) the output of *tolerance.landscape* (see ‘*Worked example*’ below).

The heuristic algorithm calculates iteratively the change in survival following a 1-min exposure to each T . In each iteration k , the algorithm shifts the survival curve based on the next T employing Eqn S2, estimates the next change in S following a 1-min exposure and adds this value to S at $\tau - 1$ (Eqn. 8). This corresponds to $S'(k \cdot \Delta\tau)$ varying with $T(k \cdot \Delta\tau)$ where $\Delta\tau = 1$ min, calculated with linear interpolation between two contiguous times in S . Importantly, because the thermal shift can accommodate any temperature value (Eqn. S2) and the vector describing S is discrete with 1000 bins, each iteration generates a mismatch between the 1-min time bins at $\tau - 1$ and S following the shift. The exact S at τ is estimated with linear interpolation and the drop in survival during the next 1-min is then calculated.

With this approach, the survival curve with ramping is built iteratively minute by minute, and in each iteration we include an interpolation component that makes the discrete S curve behave as if it were continuous. Because the linear interpolation is performed between contiguous points separated by 0.1% survival (see above), any noise due to the interpolation procedure will be minimal when compared to the uncertainty associated with estimating $S_T(\tau)$ from empirical data.

Usage (note that `tl` is defined above):

```
dl <- dynamic.landscape(ambient temperature,tl)
```

Output:

`dl$time`: The time in minutes.

`dl$ta`: The ambient temperature.

`dl$alive`: The predicted fraction of individuals alive (%).

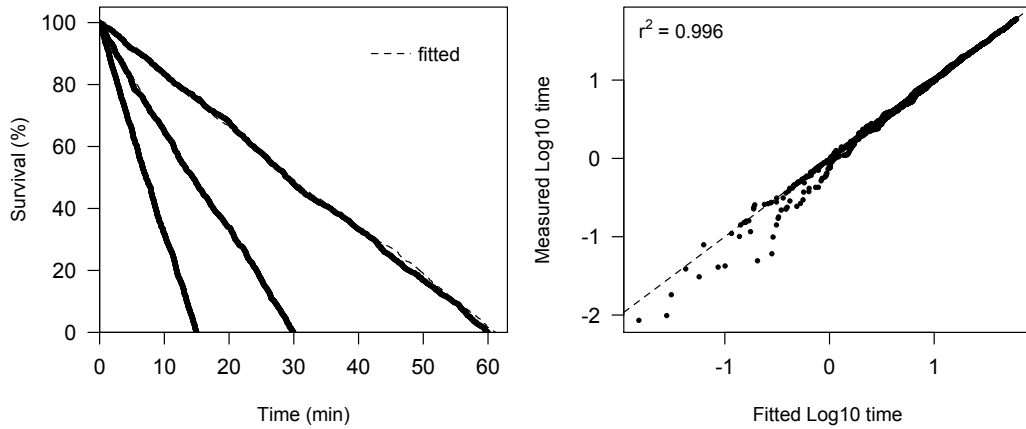
Worked examples

Example 1. Below we provide the R script for a simulated experiment in which 1000 individuals are subjected to constant 30, 31 and 32 °C, and survival is assumed to decrease linearly to 0% after 1 h, 30 min and 15 min, respectively (i.e., a 1 °C increase decreases survival times by ½). This is accomplished by sampling lethal times from a uniform distribution within these boundaries. Then, we predict $S_T(\tau)$ for a warming experiment where the population is exposed to each of these temperatures for 10 min. The plots shown in the next page are provided automatically with the two functions and the script below, and mortality can be calculated as $100 - \% \text{ survival}$:

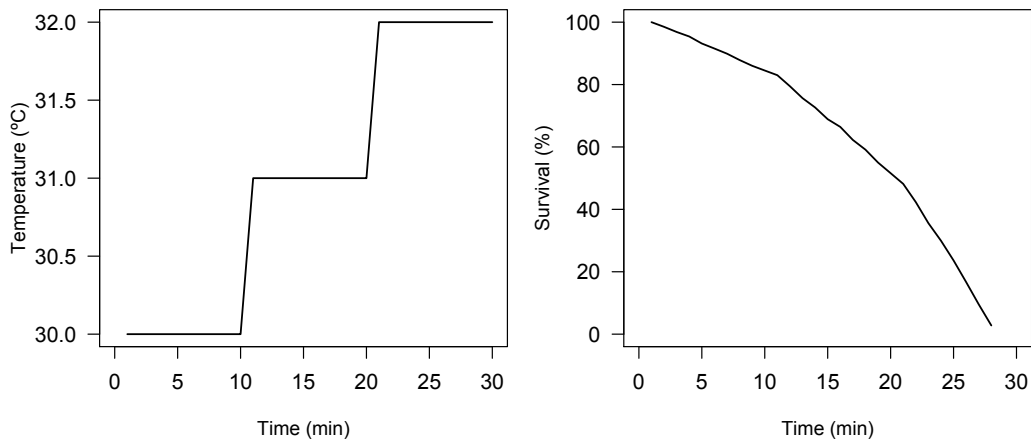
```
> set.seed(1)
> ind <- 1000
> static.temp <- rep(30:32,each=ind)
> time <- c(runif(ind,0,60),runif(ind,0,30),runif(ind,0,15))
>
> tl <- tolerance.landscape(static.temp,time)
>
> variable.temp <- rep(30:32,each=10)
> dl <- dynamic.landscape(variable.temp,tl)
```

Output of *tolerance.landscape*

Data points are ‘empirical values’ and the dotted line the fitted model.



Output of *dynamic.landscape*

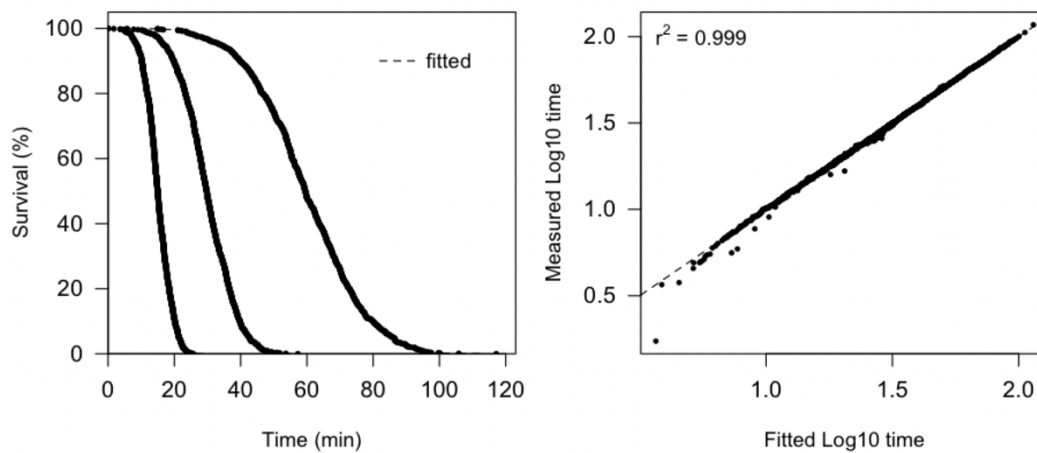


Note the breakpoints in $S_T(\tau)$ at 11 and 21 min, expected from the linear survival curves above combined with the stepwise increase in ambient temperature below.

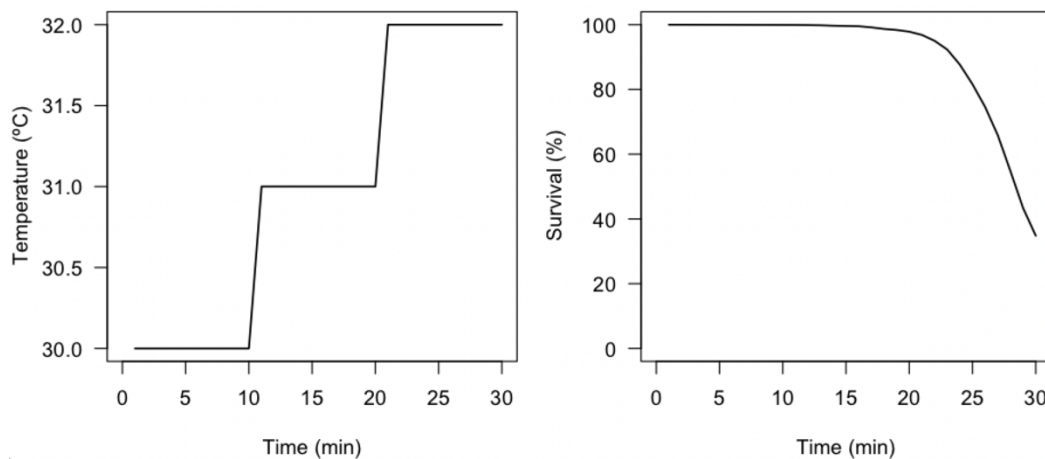
Example 2. The same simulation with different lethal times during static assays, which are now normally distributed with an average $\tau = 1$ h, 30 min and 15 min and a standard deviation corresponding to $\frac{1}{4}$ th of the mean. This distribution gives rise to sigmoidal $S_T(\tau)$.

```
> set.seed(1)
> ind <- 1000
> static.temp <- rep(30:32,each=ind)
> time <- c(rnorm(ind,60,60/4), rnorm(ind,30,30/4), rnorm(ind,15,15/4))
>
> tl <- tolerance.landscape(static.temp,time)
>
> variable.temp <- rep(30:32,each=10)
> dl <- dynamic.landscape(variable.temp,tl)
```

Output of *tolerance.landscape*



Output of *dynamic.landscape*



Validation with empirical datasets

Static assays

Below we show how the procedure works for the curves that we reported for *Drosophila subobscura* elsewhere (23). We employ this dataset because sample sizes to estimate survival probability curves are, to our knowledge, the largest currently available in the literature ($n = 1,136$ and $1,182$ for warm- and cold-acclimated flies, respectively), corresponding to roughly 300 flies per treatment measured at each test temperature ($T = 35, 36, 37$ and 38 °C). Such a large sample size allows a more accurate characterization of the survival probability curves at each temperature and illustrates how the general shape of these curves remains constant across temperatures, which is a general assumption of our proposed approach. For completeness, note that the original survival curves are shown in Fig. 5 of ref. 23.

Figure S1: Castañeda *et al.* data set (23): warm-acclimated flies.

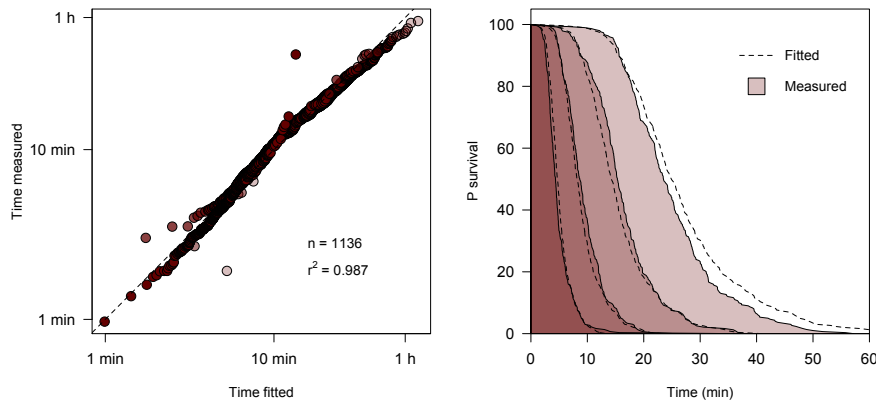
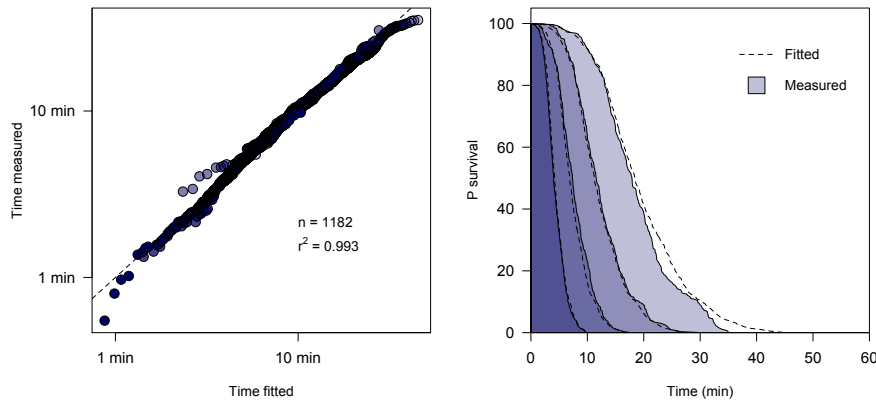
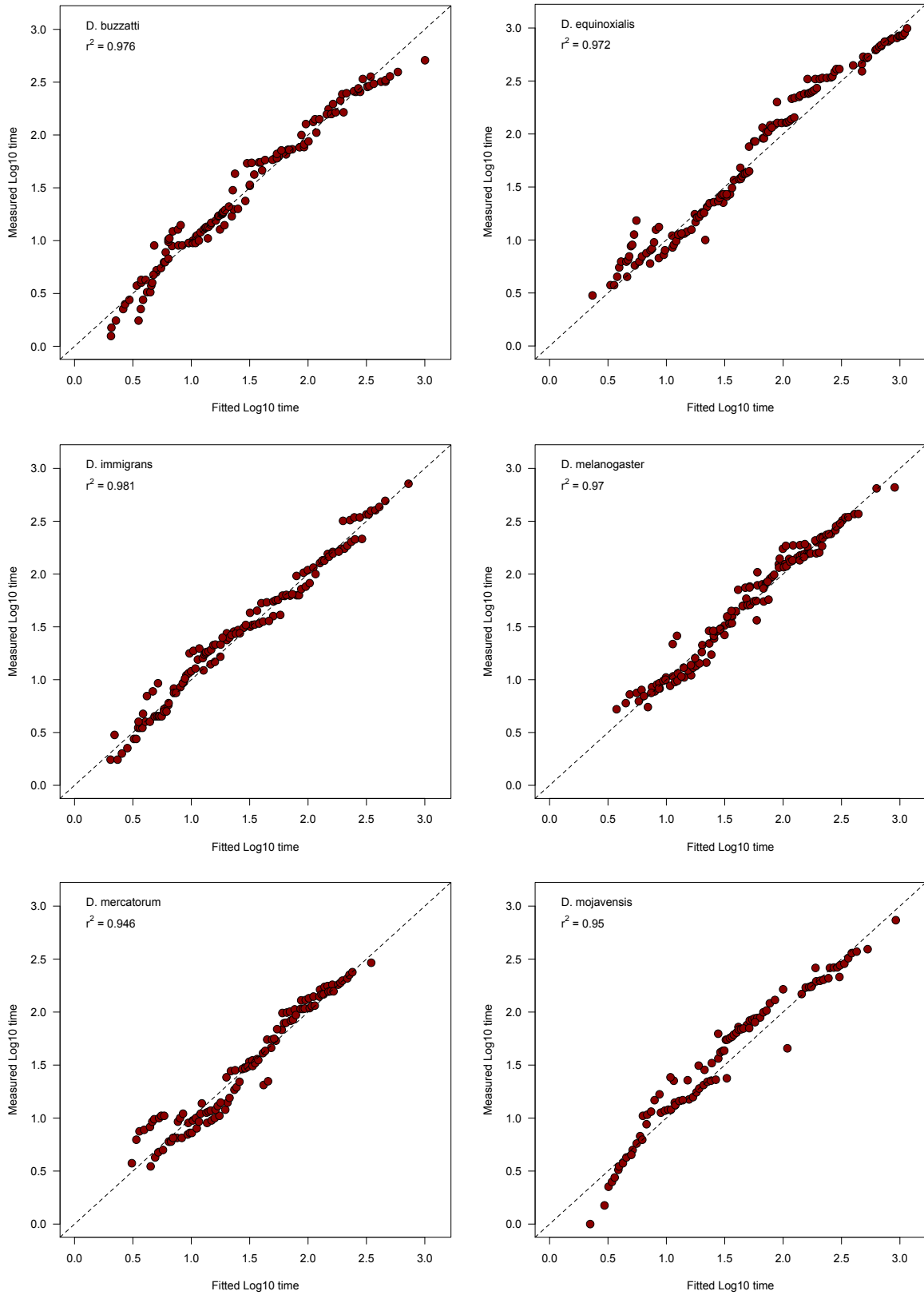


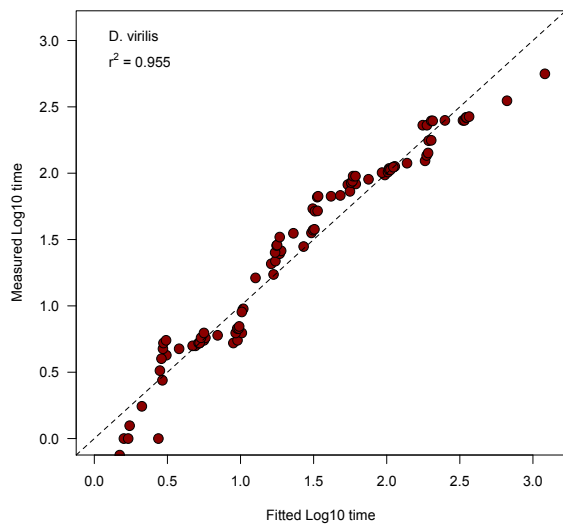
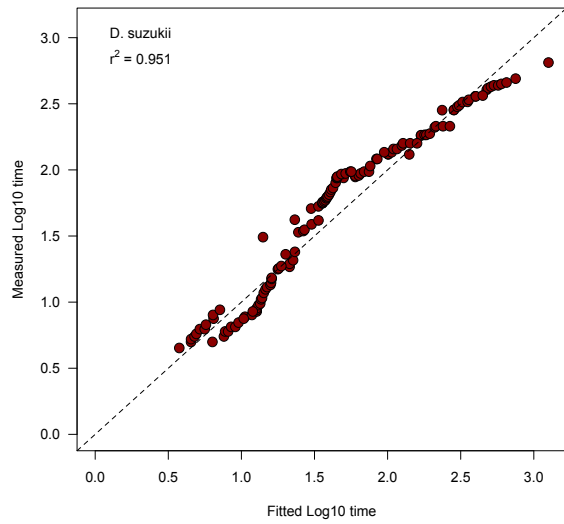
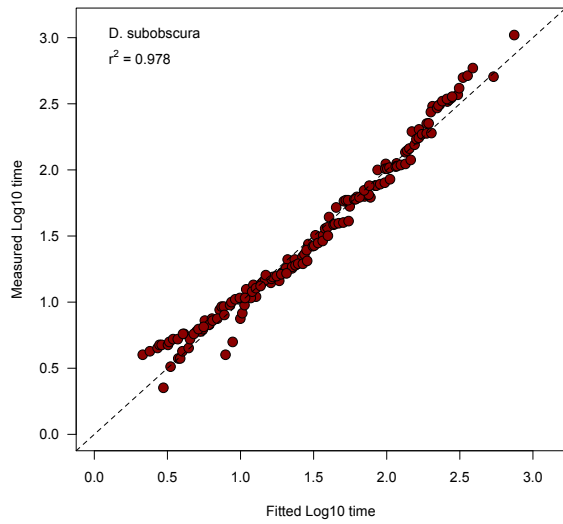
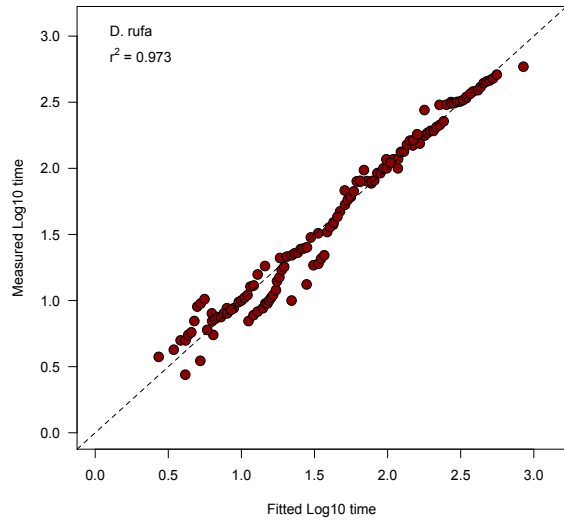
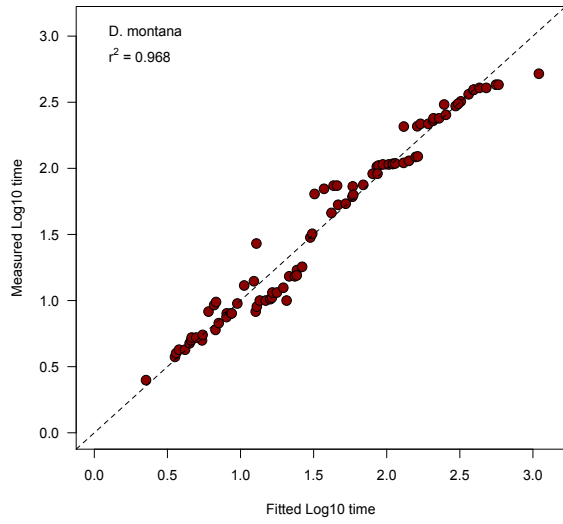
Figure S2: Castañeda *et al.* data set (23): cold-acclimated flies.



We now apply the analytical procedure to the data sets in ref. (17).

Figure S3: Jørgensen *et al.* data set (17).

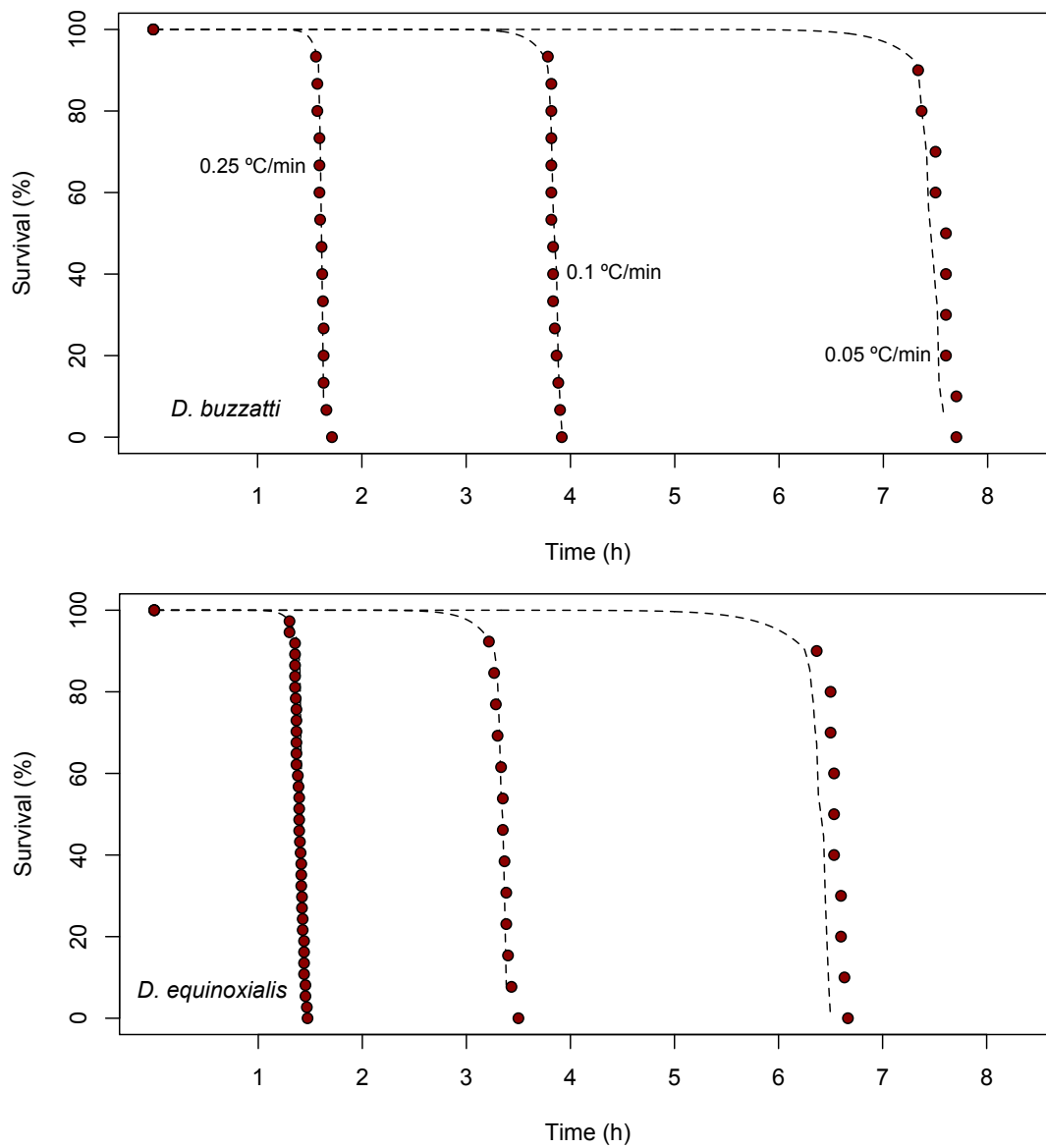


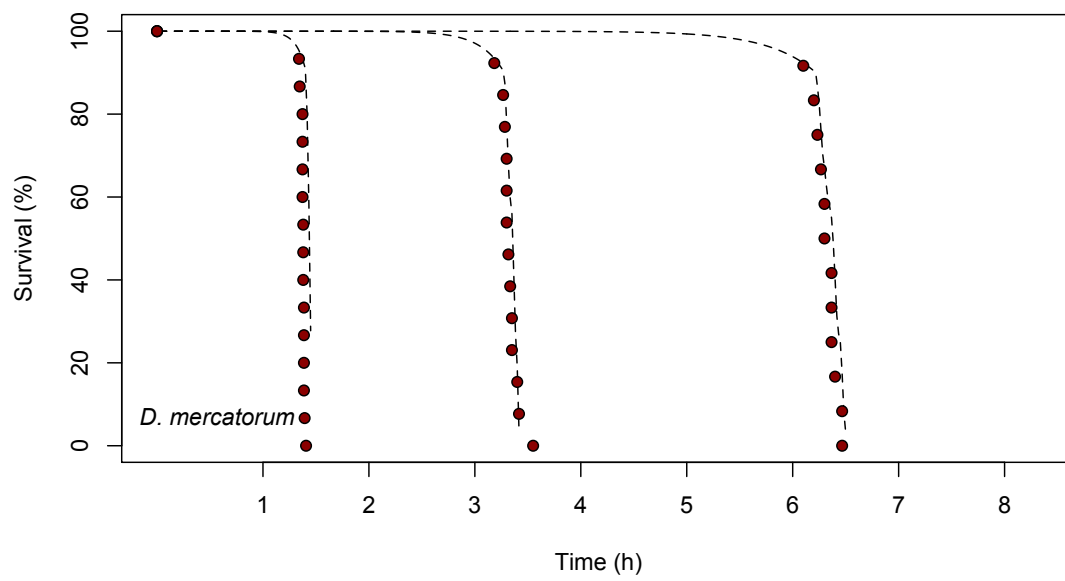
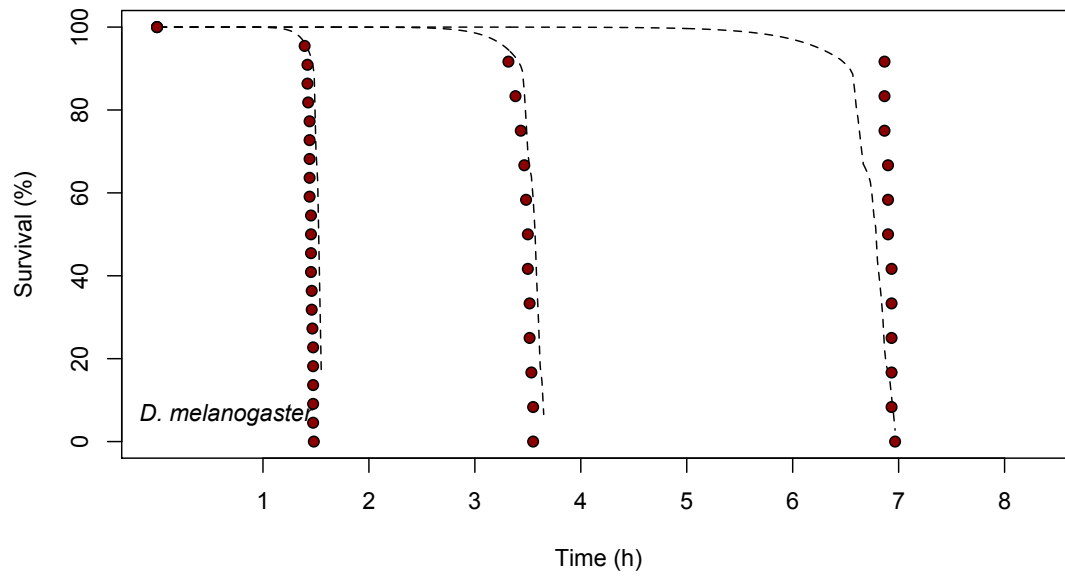
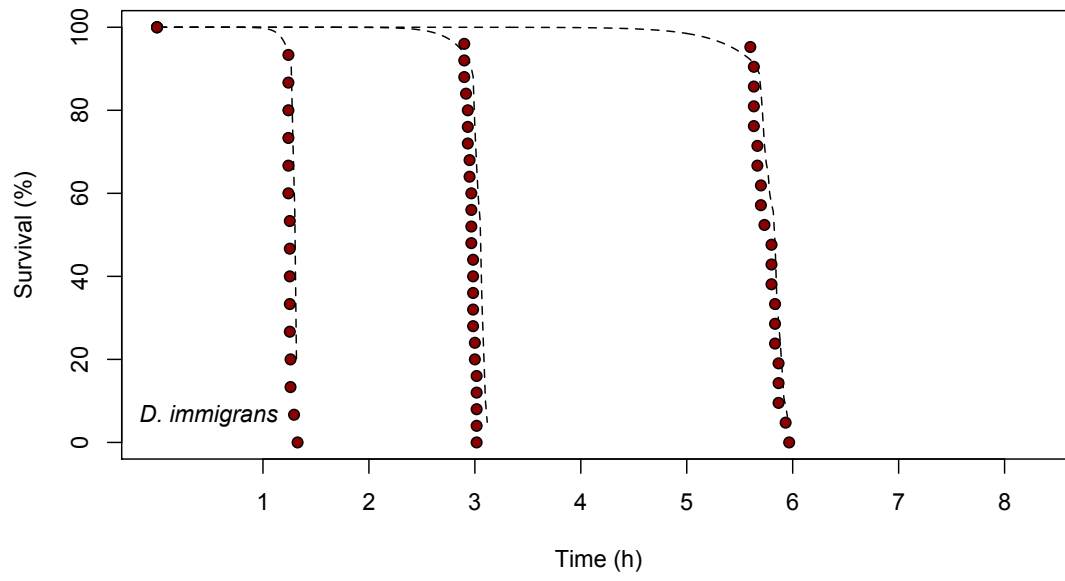


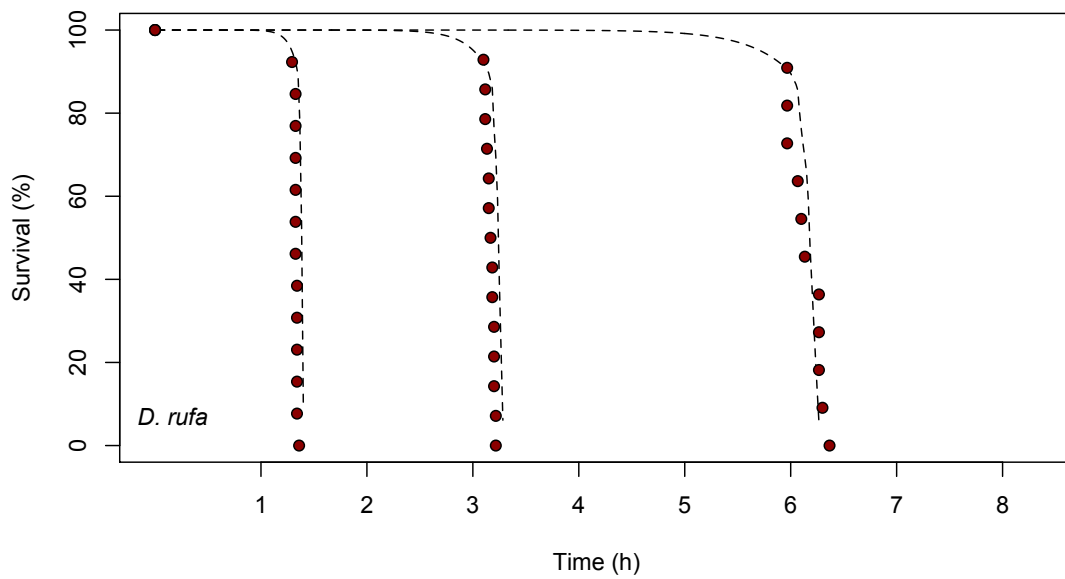
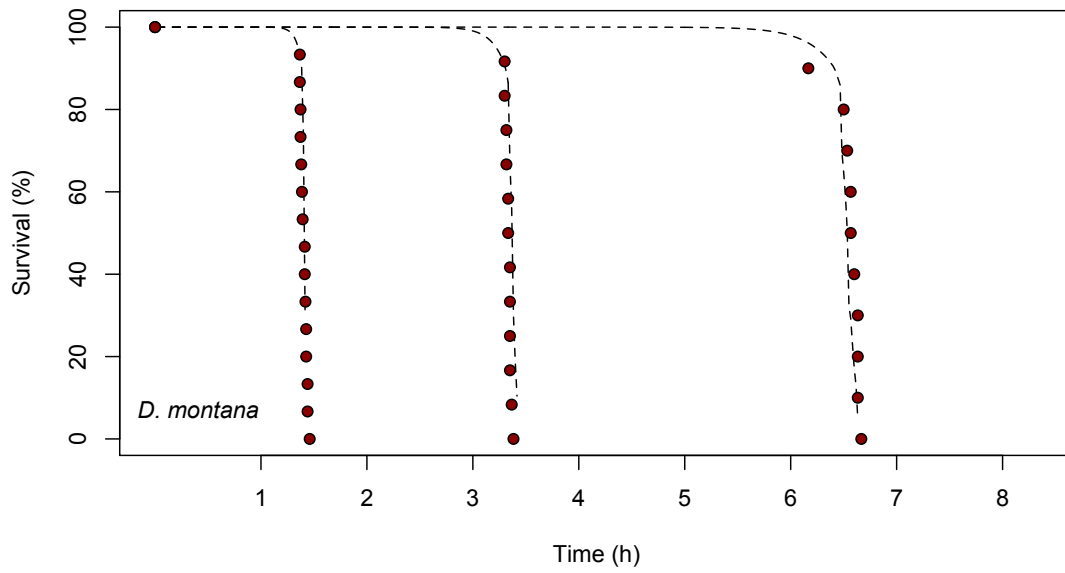
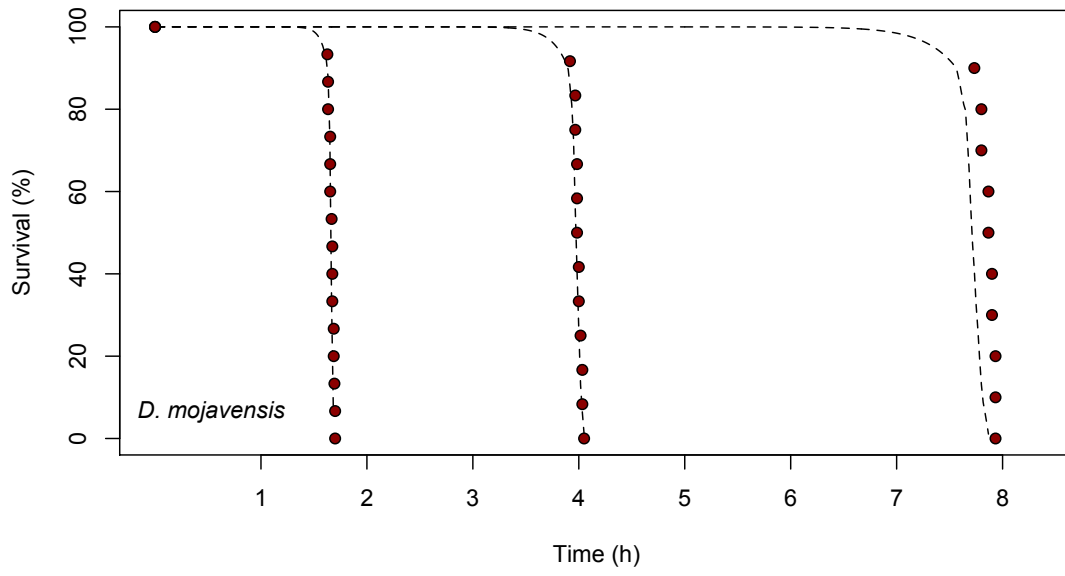
Dynamic assays

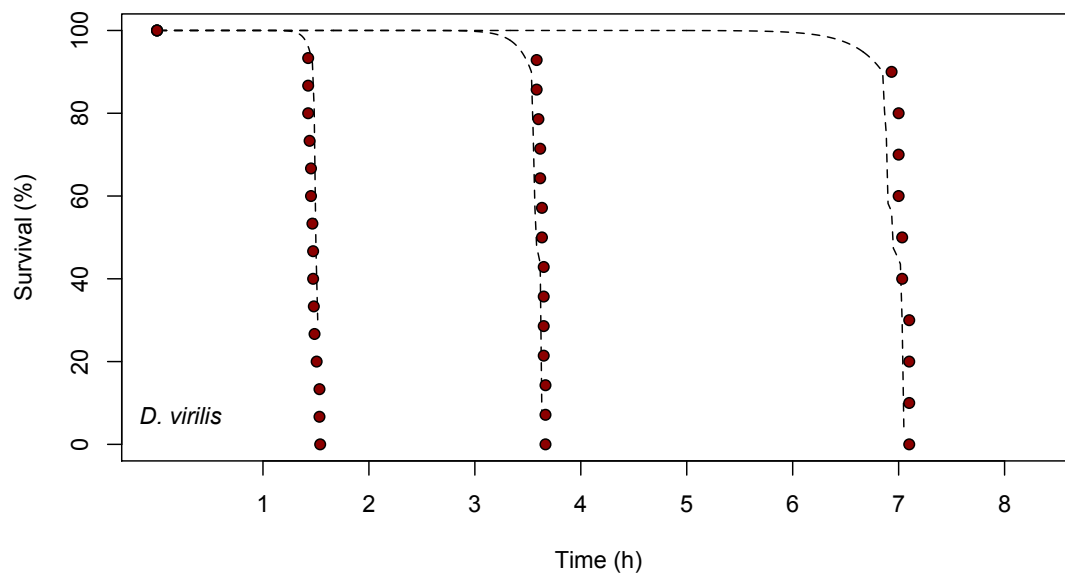
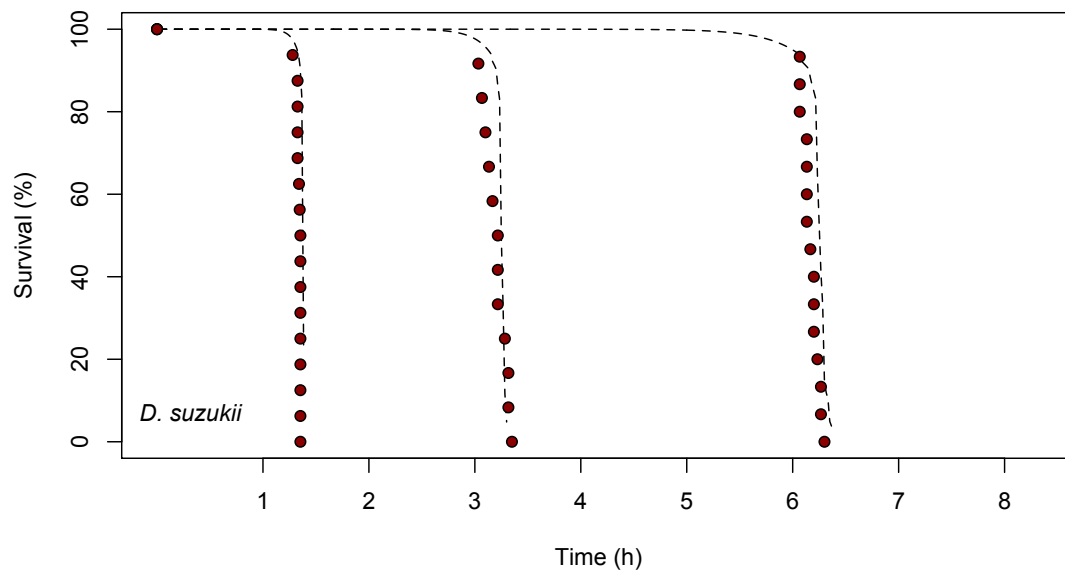
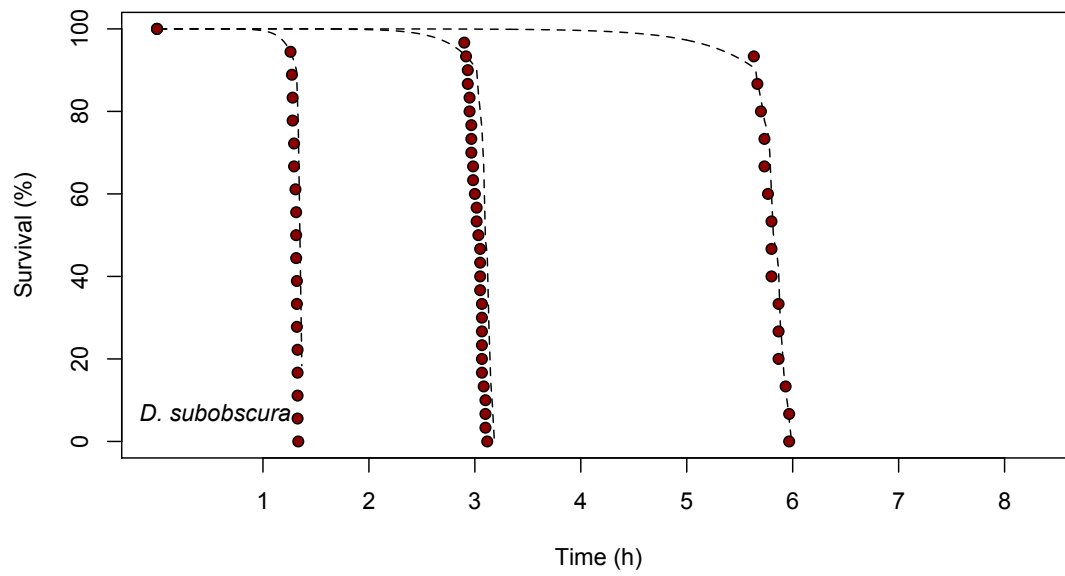
We contrasted the predictions of our model, based on static assays (Fig. S3), against empirical survival curves described for the same 11 species of *Drosophila* under three ramping conditions (17): starting at 19 °C, temperatures increased at rates of 0.05, 0.1 and 0.25 °C/min. The empirical data points are contrasted against predictions below:

Figure S4: Predicting mortality in dynamic (ramping) assays.









Thermal mortality and global warming

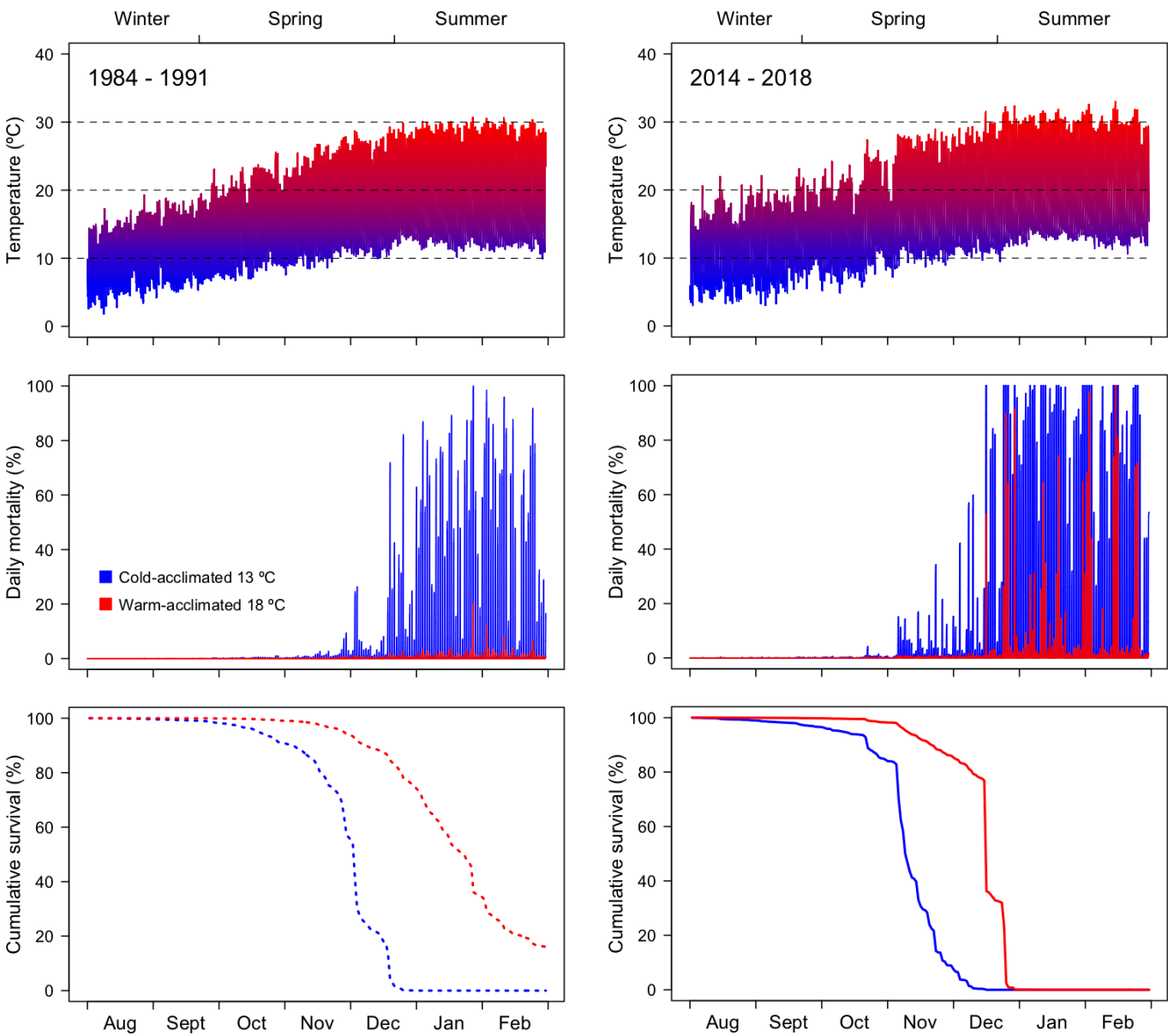
Predictions of thermal mortality were based on hourly temperature records between 2014 – 2018, whereas the relative abundance of *D. subobscura* were monitored in a longitudinal study between 1984 – 1991. To ensure that long-term differences in temperature did not alter our general conclusions, and to quantify the potential impact of global warming, we repeated analyses with hourly temperatures for both periods (data for 1984 – 1991 from <https://climatologia.meteochile.gob.cl/application/historicos/datosDescarga/330021>). Average differences in daily maximum temperatures are listed below (Table S1). Analyses suggest that warming temperatures during this 30-year period has increased mortality rates earlier in the Spring and Summer seasons, shifting the cumulative mortality and the expected abundance peak towards lower temperatures (Fig. S5).

Table S1. Historical temperature records (mean $\pm$ SD) for the periods included in this study

	Mean maximum temperature (°C)		ΔT (°C) [§]
	1984 – 1991	2014 – 2018	
January	30.0 $\pm$ 1.0	31.0 $\pm$ 1.4	1.0 $\pm$ 1.7
February	29.7 $\pm$ 0.8	30.3 $\pm$ 0.8	0.6 $\pm$ 1.1
March	27.2 $\pm$ 0.7	28.4 $\pm$ 1.0	1.2 $\pm$ 1.2
April	23.3 $\pm$ 1.1	23.6 $\pm$ 2.0	0.3 $\pm$ 2.3
May	17.7 $\pm$ 1.5	18.9 $\pm$ 1.4	1.2 $\pm$ 2.0
June	15.5 $\pm$ 2.3	16.0 $\pm$ 1.9	0.5 $\pm$ 3.0
July	14.8 $\pm$ 1.2	15.6 $\pm$ 0.3	0.8 $\pm$ 1.2
August	16.1 $\pm$ 1.2	17.5 $\pm$ 1.1	1.4 $\pm$ 1.6
September	18.3 $\pm$ 0.8	19.8 $\pm$ 2.1	1.5 $\pm$ 2.2
October	22.6 $\pm$ 1.0	23.0 $\pm$ 1.6	0.4 $\pm$ 1.9
November	26.0 $\pm$ 1.6	26.6 $\pm$ 1.5	0.6 $\pm$ 2.2
December	28.8 $\pm$ 0.8	29.3 $\pm$ 0.8	0.5 $\pm$ 1.1

[§] SD estimated as $\sqrt{\sigma_1^2 + \sigma_2^2}$ for variances that are independent.

Figure S5: Shift in mortality rates during 30 years of global warming. Note the increase in daily temperatures and mortality. Cumulative mortalities in the lower panels are shown side-by-side in Fig. 3c.



References and Notes

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